untitled

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```
from collections import deque

M_TOTAL = 3
C_TOTAL = 3
BOAT = 2

# state = (missionaries on left, cannibals on left, boat side: 1 = left, 0 = right)

def valid(m, c):
    if not (0 <= m <= M_TOTAL and 0 <= c <= C_TOTAL):
        return False
    mr, cr = M_TOTAL - m, C_TOTAL - c
    if m > 0 and c > m:
        return False
    if mr > 0 and cr > mr:
        return False
    return True

def successors(state):
    m, c, boat = state
    sign = -1 if boat == 1 else 1
    for dm in range(BOAT + 1):
        for dc in range(BOAT + 1 - dm):
            if 1 <= dm + dc <= BOAT:
                nm, nc = m + sign * dm, c + sign * dc
                if valid(nm, nc):
                    yield (nm, nc, 1 - boat), (dm, dc)

def solve():
    start = (M_TOTAL, C_TOTAL, 1)
    goal = (0, 0, 0)
    parent = {start: None}
    queue = deque([start])
    while queue:
        state = queue.popleft()
        if state == goal:
            path = []
            while parent[state] is not None:
                prev, move = parent[state]
                path.append((move, state))
                state = prev
            return list(reversed(path))
        for nxt, move in successors(state):
            if nxt not in parent:
                parent[nxt] = (state, move)
                queue.append(nxt)
    return None
```

Ln: 56 Col: 48

26°C Light rain

ENG IN 09:28 25-10-2026

```
Start: left bank = 3M 3C, right bank = 0M 0C, boat on LEFT

Step  1: send 0M 2C left -> right  | left = 3M 1C | right = 0M 2C
Step  2: send 0M 1C right -> left  | left = 3M 2C | right = 0M 1C
Step  3: send 0M 2C left -> right  | left = 3M 0C | right = 0M 3C
Step  4: send 0M 1C right -> left  | left = 3M 1C | right = 0M 2C
Step  5: send 2M 0C left -> right  | left = 1M 1C | right = 2M 2C
Step  6: send 1M 1C right -> left  | left = 2M 2C | right = 1M 1C
Step  7: send 2M 0C left -> right  | left = 0M 2C | right = 3M 1C
Step  8: send 0M 1C right -> left  | left = 0M 3C | right = 3M 0C
Step  9: send 0M 2C left -> right  | left = 0M 1C | right = 3M 2C
Step 10: send 0M 1C right -> left  | left = 0M 2C | right = 3M 1C
Step 11: send 0M 2C left -> right  | left = 0M 0C | right = 3M 3C

Solved in 11 crossings.
```